Figure 5.1 Modern developers have to consider several consumers of their applications. As well as traditional users with web browsers, these users could be single-page applications, mobile applications, or other apps.

Client-side single-page applications (SPAs) have become popular in recent years with the development of frameworks such as Angular, React, and Vue. These frameworks typically use JavaScript running in a web browser to generate the HTML that users see and interact with. The server sends this initial JavaScript to the browser when the user first reaches the app. The user's browser loads the JavaScript and initializes the SPA before loading any application data from the server.

NOTE Blazor WebAssembly is an exciting new SPA framework. Blazor lets you write an SPA that runs in the browser like other SPAs, but it uses C# and Razor templates instead of JavaScript by using the new web standard, WebAssembly. I don't cover Blazor in this book, so to find out more, I recommend *Blazor in Action*, by Chris Sainty (Manning, 2022).

Once the SPA is loaded in the browser, communication with a server still occurs over HTTP, but instead of sending HTML directly to the browser in response to requests, the server-side application sends data—normally, in the ubiquitous JavaScript Object Notation (JSON) format—to the client-side application. Then the SPA parses the data and generates the appropriate HTML to show to a user, as shown in figure 5.2. The server-side application endpoint that the client communicates with is sometimes called an *HTTP API*, a *JSON API*, or a *REST API*, depending on the specifics of the API's design.